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Virus inactivation and protein recovery in a novel ultraviolet-C reactor

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Vox Sanguinis

Background and Objectives Ultraviolet-C (UVC) irradiation is a viral-inactivation method that was dismissed by many plasma fractionators as a result of the potential for protein damage and the difficulty in delivering uniform doses. A reactor with novel spiral flow hydraulic mixing was recently designed for uniform and controlled UVC treatment. The objective of this study was to investigate virus inactivation and protein recovery after treatment through the new reactor.

Materials and Methods Virus- and mock-spiked Alpha$_1$-proteinase inhibitor (Alpha$_1$-PI) solutions were treated with UVC. The virus samples were assayed for residual infectivity and amplified by the polymerase chain reaction (PCR). The mock-spiked samples were assayed for protein integrity.

Results Greater than 4 $\log_{10}$ of all test viruses were inactivated, regardless of the type of nucleic acid or presence of an envelope. Unlike previous studies, viruses with the smallest genomes were found to be those most sensitive to UVC irradiation, and detection of PCR amplicons ≥ 2·0 kb was correlated to viral infectivity. Doses that achieved significant virus inactivation yielded recovery of > 90% protein activity, even in the absence of quenchers.

Conclusions The results demonstrate the effectiveness of UVC treatment, in the novel reactor, to inactivate viruses without causing significant protein damage, and confirm the utility of large PCR amplicons as markers for infectious virus.

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Introduction

Plasma-derived therapeutic proteins are parenteral biologicals that are purified on an industrial scale, and all biologicals derived from human plasma carry the risk of virus contamination. The viral safety of plasma derivatives has increased dramatically over the last 10–15 years as a result of improved donor screening and plasma-testing methods, quarantine of collected plasma, and implementation of virus-reduction steps in production processes. Virus-inactivation methods, such as dry heat treatment, pasteurization, low-pH incubation, and treatment with solvent/detergent or caprylate, have significantly reduced the risk of transmission of enveloped viruses, such as human immunodeficiency virus (HIV), hepatitis B virus (HBV) and hepatitis C virus (HCV). However, many of these methods are less effective against small, heat-resistant, non-enveloped viruses, such as hepatitis A virus (HAV) and human parvovirus B19 (B19), and therefore the need for inactivation techniques for non-enveloped viruses still remains.

Treatment with ultraviolet-C (UVC) irradiation, one of many viral-inactivation approaches under development for blood and plasma derivatives, was first introduced in 1949 [1]. Initially, DNA was regarded as the only molecular target of UVC [2], but later studies showed that UVC treatment

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could also indirectly damage proteins through the generation of reactive chemical species [3]. Many early UVC devices were laminar flow reactors that suffered from the poor transmissibility and low effective depth penetration of UVC irradiation. As radiation intensity decreases exponentially with increasing distance from an irradiation source, liquid media were irradiated as thin films, and for prolonged time periods, to ensure viral inactivation. Commercial-scale laminar flow reactors never became popular for biological products because maintaining film thinness required extremely large irradiation surfaces and extending the exposure times increased the potential for protein damage. Scavengers or quenchers of free radicals were added to selectively protect proteins against UVC [4,5], but additional downstream steps were then necessary to remove these chemicals, increasing process time and decreasing yields.

A new generation of continuous-flow UVC reactors has been designed by Bayer Technology Services to eliminate the need for laminar flow thin films. In the Bayer Technology Services UV reactor, novel hydraulic spiral flow along an irradiation source induces highly efficient mixing in a fluid stream, so high doses of UVC irradiation can be delivered evenly and uniformly throughout the solution. Thus, the required residence times in the irradiation chamber are extremely short and UVC treatment is controllable. The objective of this study was to investigate virus inactivation and plasma protein recovery by UVC treatment in the new-generation reactor in the absence of quenchers. During these studies, $Alpha_1$-proteinase inhibitor ($Alpha_1$-PI), a plasma protein that inhibits neutrophil elastase [6], was selected as the target protein owing to the presence of UV-absorbing amino acids near the enzyme's reactive centre [7,8]. If UVC (254 nm) treatment damaged $Alpha_1$-PI, the UV-absorbing amino acids would probably be involved and the UV-induced damage could be easily detected by a decrease in biological activity.

Materials and methods

Viruses and cells

The viruses used in this study – B19, porcine parvovirus (PPV), simian virus 40 (SV40), HAV, Sindbis virus (Sindbis), reovirus type 3 (Reo) and adenovirus type 5 (Adeno), were selected according to the Committee for Proprietary Medicinal Products (CPMP) guidelines on virus studies [9]. B19 and HAV were used as clinically relevant viruses for plasma-derived products, while Sindbis and PPV were used as surrogates for HCV and B19, respectively. SV40, HAV, B19 and PPV represented small, non-enveloped viruses with high resistance, while Reo and Adeno modelled large non-enveloped viruses with medium resistance to physico-chemical methods of inactivation.

Human plasma, containing B19 at 10^{11} IU/ml, was obtained from the Nucleic Acid Testing Facility, Pre-clinical Research and Pathogen Safety, Bayer Healthcare (Raleigh, NC). PPV and PT cells, which were used to propagate and assay PPV, were obtained from the Biological Research Faculty and Facility (Ijamsville, MD). The remaining viruses and cells were obtained from the American Type Culture Collection (Manassas, VA). SV40, HAV and Reo were grown and assayed in BSC-1 cells, while 293 and BHK-21 cells were used for Adeno and Sindbis virus, respectively. Cells were grown in Eagle's minimal essential medium containing 0·1 mM non-essential amino acids, 10 mM *N*-(2-hydroxyethyl)-piperazine-*N′*-(2-ethanesulphonic acid), gentamicin and fungizone (MEM + NHGF), supplemented with 7·5% fetal bovine serum (FBS). All viruses, except PPV and HAV, were propagated by infecting the appropriate cells at a low multiplicity of infection, adding MEM + NHGF + 2% FBS and then incubating the cells at 37 °C in a CO_2 incubator until advanced cytopathic effects (CPE) were observed (4–7 days postinfection). For PPV propagation, the FBS content of the medium was increased to 7·5%. For HAV propagation, the temperature of incubation was decreased to 35 °C and the length of incubation was extended to 8–10 days. All virus-infected cells were frozen and thawed three times to release virus, the infected cell lysates were centrifuged at low speed (1700 ***g***, 5 °C, 15 min) to remove cell debris and the clarified supernatants were collected for use as the virus-spike in experiments.

Plasmids

B19 plasmid, PYT-104-C (GenBank accession no.: M13178), was obtained from the Nucleic Acid Testing Facility, Pre-clinical Research and Pathogen Safety (Bayer Healthcare). PPV plasmid, pN2Δ, was a kind gift from P. Tijssen (Institute Armand-Frappier, University of Quebec, Laval, Canada). The pN2Δ plasmid contains the complete NADL-2 genome (GenBank accession no.: L23427) in a pUC19 background and was propagated and purified by Aldevron (Fargo, ND).

Bayer Technology Services UV Reactor

All UVC experiments were performed in the Bayer Technology Services UV reactor (US patent application: US 2003/0049809A1), equipped with a low-pressure mercury lamp, model CPQ-7731, from UV-und Labortechnik (Glashutten, Germany). A tubular poly(tetrafluorethylene) conduit, which spirals around a quartz tube with a concentric UVC source (254 nm), forms the irradiation chamber of the reactor. The irradiation surface area is 0·0246 m^2 and the volume of the irradiation chamber is 24 ml. As fluid streams move spirally along the lamp, secondary circulating flows (Dean vortices) are generated that provide highly efficient mixing and allow for uniform and controllable irradiation of the entire volume.

Determination of UV fluency

UV dosage is typically described in units of UV fluency (W s/cm^2 = J/cm^2) and is dependent on the optical density of the test solution, average irradiance emitted by the lamp and residence time in the irradiation chamber. Actinometric experiments with potassium ferroxalate [10,11] confirmed information from the lamp manufacturer and showed that the lamp intensity was 0·0055 W/cm^2. Spectrophotometric readings, at 254 nm, showed that the optical density of the Alpha$_1$-PI solution was 2·500. As efficient mixing by the generation of Dean vortices ensures that every liquid element passes through the irradiation zone, the average UV irradiance (0·00225 W/cm^2) was based on the geometry of the irradiation chamber and the law of Lambert-Beer. Multiplication of the average lamp irradiance with the residence time in the irradiation chamber yielded the average UV fluency. For example, if the residence time of the test solution in the irradiation chamber was 4·1 s (500 ml/min), then the UV fluency = 0·00225 W/cm^2 × 4·1 s = 0·009 W s/cm^2 = 0·009 J/cm^2.

UVC Treatment of Alpha$_1$-PI

Prolastin®, a sterile stable lyophilized preparation of purified human Alpha$_1$-PI, also known as α_1-antitrypsin, was obtained from the Bayer Healthcare plasma fractionation facility (Clayton, NC). For the virus experiments, lyophilized Alpha$_1$-PI was reconstituted in buffer (0·1 M sodium chloride, 0·02 M sodium phosphate, pH 7), and 1900 ml of the 0·8% protein solution was spiked with 100 ml of virus. A 20-ml aliquot was removed, titrated for virus as the INITIAL sample and then saved for polymerase chain reaction (PCR) analyses. The remaining solution was passed through the Bayer Technology Services UV reactor at different flow rates to increase or decrease UVC fluencies. Two or more 15-ml aliquots of the UVC-treated test solutions were collected at various fluencies (0·007–0·431 J/cm^2), titrated for virus as the UVC-TREATED samples, and then saved for PCR analyses. Characterization samples were also generated to measure Alpha$_1$-PI activity and monomer levels. As the bioanalytical assays to measure Alpha$_1$-PI were performed outside the virus containment area, characterization runs were performed exactly like the virus runs, except that the virus spike was replaced with buffer.

Determination of Alpha$_1$-PI activity

Alpha$_1$-PI activity was determined by inhibition of porcine pancreatic elastase using a chromogenic substrate [succinyl-(alanine)$_3$-p-nitro-anillide] [12]. Briefly, test samples and controls were diluted in buffer and incubated with porcine pancreatic elastase. Chromogenic substrate was added and hydrolysed by elastase that was not inhibited by Alpha$_1$-PI. Thus, the amount of hydrolysis, as quantified by measuring the absorbance at 405 nm, was inversely proportional to the concentration of Alpha$_1$-PI present. All assays were performed using an automated coagulation analyser, MLA Electra 1600C (Medical Laboratory Automation, Inc., Pleasantville, NY).

Determination of Alpha$_1$-PI aggregate levels by size-exclusion high-performance liquid chromatography

The molecular-weight distribution of Alpha$_1$-PI was determined by size-exclusion high-performance liquid chromatography (SEC-HPLC), using a TSK G3000 type SW column (Toyo Soda Corporation, Japan) and an isocratic HPLC system consisting of a Hewlett Packard 1050 or 1100 series autosampler, pump and variable wavelength detector. Quantification was based on ultraviolet absorbance at 280 nm, and the results were expressed as relative peak areas.

Heating of 5% albumin solutions

Human albumin, 5%, obtained from the Bayer Healthcare plasma fractionation facility (Clayton, NC), was aliquoted into tubes (9·5 ml per tube) and then each tube was spiked with 0·5 ml of PPV to yield final concentrations of 10^5, 10^4, 10^3 and 10^2 tissue culture infective dose 50% (TCID$_{50}$) per ml. One set was heated at 61 °C for 20 h, while a duplicate set was held at 5 °C as unheated controls. Tubes of mock-spiked albumin were also incubated at both temperatures and served as negative controls. After treatment, UNHEATED and HEATED samples were removed for virus titration and PCR analyses.

Quantification of virus

Virus titrations were performed in 96-well microtiter plates seeded with the appropriate cells, as previously described [13]. Briefly, serial half-log dilutions of virus test samples were made in Hank's balanced salt solution (HBSS) and 0·1-ml aliquots of each sample dilution were added to the microtitre plate (eight replicate wells per dilution). After virus adsorption at 37 °C, the inocula were removed and replaced with fresh medium. PPV assays were performed with MEM + NHGF + 7·5% FBS, while all other virus assays were carried out in MEM + NHGF + 2% FBS. Except when assaying for HAV, all virus-infected cells were incubated for 7 days at 37 °C and then examined under a microscope for the presence of viral CPE. HAV-infected cells were incubated at 35 °C for 21 days before microscopic examination. Virus titres were determined by TCID$_{50}$, using the method of Spearman-Karber [14]. The detection limits of the virus assays were determined by cytotoxicity and viral interference assays. Alpha$_1$-PI was not toxic to any of the indicator cells at any of the dilutions tested and did not interfere with virus infection (data not shown).

Based on these results, the detection limit of virus in Alpha$_1$-PI was set at ≤ 0·7 log $TCID_{50}$/ml.

Calculation of virus-reduction factors

The effectiveness of any virus-reduction method to inactivate virus can be expressed as the virus-reduction factor and is determined by comparing the virus load in the input with that of the output [9]. For the UVC and wet heat experiments, the volumes before and after treatment did not change, so the $\log_{10}$ virus-reduction factors were calculated as the difference in $\log_{10}$ virus titres between the INITIAL and UVC-TREATED, or between the UNHEATED and HEATED, samples.

Primers for PCR analysis

The PPV primers used to amplify the 0·3-kb amplicon were ppv-3-antisense (1328–1350; 5′-TGGTAGGATGCTTGGTTTTGCC-3′) and ppv-5-sense (1012–1037; 5′-GAAGGCGAGAGACACT-TAGTCAGTC-3′), while the PPV primers used to amplify the 0·7-kb amplicon were ppv-4-antisense (1656–1678; 5′-TTGTTTGACCTGAACATATGGC-3′) and ppv-5-sense. For the PPV 1·2-kb amplicon, the primers ppv-1-antisense (2182–2205; 5′-TGGACTTGGTGTCCGTATTGCTG-3′) and ppv-5-sense were used, and for the PPV 2·0-kb amplicon, the primers ppv-1-antisense and ppv-6-sense (231–251: 5′-CTGCTTCA-GACTGCACTTCGC-3′) were used. The B19 primers used to amplify the 0·4-kb amplicon were PVB-sense (2714–2738; 5′-CCCATGCCTTATCATCCAGTAGCA-3′) and PVH-antisense (3130–3106; 5′-GCTGGCTTCTGCAGAATTAACTGA-3′), while the B19 primers for the 2·2-kb amplicon were PVD-sense (961–984; 5′-GCCAAGAAACCCCGCATTACCAC-3′) and PVH-antisense.

PCR analysis of viral DNA

DNA was extracted from test samples using the Qiagen QIAamp® UltraSens™ Virus Kit (Valencia, CA; Catalogue no. 53704). Hot-start PCR was performed in a GeneAmp® PCR System 9700 (Perkin-Elmer Cetus, Foster City, CA), using the Qiagen HotStarTaq™ Master Mix Kit (Catalogue no. 203203). Five microlitres of extracted DNA was added to a final volume of 50 µl containing 1 × HotStarTaq master mix (HotStarTaq master mix, Valencia, CA) and 0·2 nM of each sequence-specific primer pair. Amplification was carried out as follows: an initial activation step at 95 °C for 15 min, followed by 30–35 cycles at 94 °C for 30 s, 60 °C for 30 s and 72 °C for 2 min; and a final step at 72 °C for 10 min with a 4 °C hold. A 20-µl sample of PCR product was analysed by electrophoresis for ≈ 1 h at 140 V, on a 1·0–1·5% agarose gel, in 1 × TAE (40 mM Tris-acetate, 1 mM EDTA). DNA was visualized by UV irradiation after staining with ethidium bromide (0·5 µg/ml in water).

Table 1 Effect of ultraviolet C (UVC) treatment on Alpha$_1$-proteinase (Alpha$_1$-PI) monomer and potency levels

UVC fluency (J/cm^2)	Monomer[a]		Potency[a]	
	% Monomer	% Recovery	mg/ml	% Recovery
0·000	77·09	100%	4·98	100%
0·009	76·94	100%	4·92	99%
0·024	77·00	100%	4·89	98%
0·049	76·64	99%	4·86	98%
0·072	76·37	99%	4·78	96%
0·144	75·13	97%	4·81	97%
0·288	71·91	93%	4·46	90%
0·431	69·43	90%	4·32	87%

[a]Average of triplicate samples.

Results

Characterization of UVC-irradiated Alpha$_1$-PI

Mock-spiked Alpha$_1$-PI was UVC irradiated by passage through the Bayer Technology Services UV reactor at different flow rates. After treatment with 0·009 J/cm^2 UVC, 99% of the initial Alpha$_1$-PI activity was recovered. An eightfold increase in fluency (0·072 J/cm^2) decreased protein activity by only 4% (Table 1). Even at the highest fluency tested, 0·43 J/cm^2, Alpha$_1$-PI recovery was 87%. The data show that the decrease in Alpha$_1$-PI monomers was minimal and corresponded to a comparable decrease in Alpha$_1$-PI potency. Treatment with 0·144 J/cm^2 UVC resulted in a 3% decrease in both monomer and protein activity recovery. Similarly, a 10% decrease in monomers was accompanied by a 13% decrease in potency after treatment with 0·431 J/cm^2 UVC.

Inactivation of viruses in Alpha$_1$-PI by UVC irradiation

Alpha$_1$-PI was spiked with virus and UVC treated. The presence of virus did not affect the UVC absorbancy of the Alpha$_1$-PI solution, as the optical density of the process stream, at 254 nm, before and after the addition of virus, did not change. Log_{10} virus-reduction factors were plotted against UVC fluencies (dose), and virus-survival curves were generated. Using the slope derived from regression analysis, the UVC fluencies required to inactivate 4 $\log_{10}$ virus (D_{4L}) were calculated.

As expected, virus inactivation by UVC treatment increased with increasing fluencies (Fig. 1a). Among the viruses tested, the 5-kb double-stranded DNA-containing SV40 was the most sensitive to UVC and was inactivated to a level lower than the detection limits at the lowest fluency tested (0·007 J/cm^2). PPV, a single-stranded DNA virus, was inactivated by 3·4 $\log_{10}$ at the same fluency, but was undetectable after

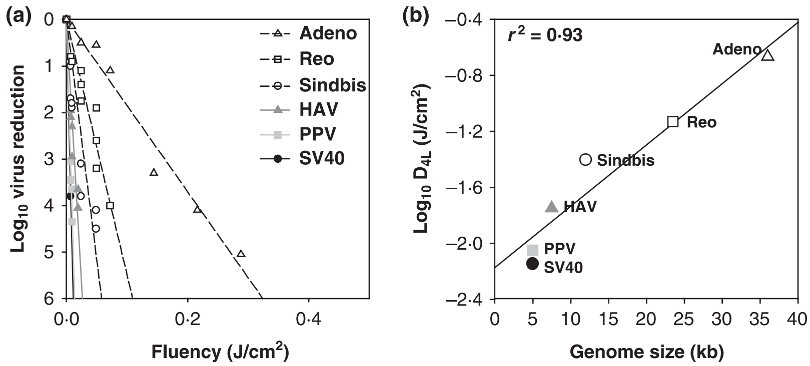

Fig. 1 (a) Virus inactivation by ultraviolet C (UVC) irradiation in the presence of Alpha$_1$-proteinase inhibitor (Alpha$_1$-PI) and (b) correlation between viral genome size and D_{4L} (fluency required for 4 $\log_{10}$ virus inactivation). Virus-spiked Alpha$_1$-PI was treated by continuous-flow UVC. Aliquots were removed at the specified fluencies and titrated for residual virus. In (a), the $\log_{10}$ virus-reduction factors from duplicate samples and replicate runs were averaged and plotted against the corresponding UVC fluencies. In (b), the D_{4L} value for each test virus was calculated, using the dose–response data from (a), and plotted against its genome size. The correlation coefficient (r^2) for the two parameters was 0·93.

Table 2 Properties and D_{4L}[a] of test viruses

Virus	Envelope	Nucleic acid[b]	Virus size (nm)	Genome size (kb)	D_{4L} (J/cm^2)
SV40	No	dsDNA	40–50	5·0	0·007
PPV	No	ssDNA	18–24	5·0	0·009
HAV	No	ssRNA	27–32	7·5	0·018
Sindbis	Yes	ssRNA	60–70	12·0	0·040
Reo	No	dsRNA	60–80	23·5	0·074
Adeno	No	dsDNA	70–90	36·0	0·216

[a]Fluency required for 4 $\log_{10}$ virus inactivation; D_{4L} values are unique to each ultraviolet (UV) system and were calculated from the data in Fig. 2.
[b]ss, single stranded; ds, double stranded.
Adeno, adenovirus type 5; HAV, hepatitis A virus; PPV, porcine parvovirus; Reo, reovirus type 3; Sindbis, Sindbis virus; SV40, simian virus 40.

treatment with 0·009 J/cm^2 UVC. The single-stranded RNA viruses were also sensitive to UVC treatment. The enveloped RNA virus, Sindbis, and the non-enveloped RNA virus, HAV, were inactivated at 4 $\log_{10}$ with doses as low as 0·040 J/cm^2 and 0·018 J/cm^2, respectively (Table 2). Large viruses, such as Reo and Adeno, were less sensitive to UVC treatment, and the D_{4L} values for these viruses ranged from 0·074 to 0·239 J/cm^2. The size of the virus genome could be correlated to the $\log_{10}$ D_{4L} values as the correlation coefficient (r^2) for the two parameters was 0·93 (Fig. 1b).

Mechanism of virus inactivation by UVC irradiation

PCR amplification was used to detect UVC-induced DNA damage. If UVC inactivation of viruses is caused by nucleic acid damage, then the presence of damaged nucleotides in the template DNA should interfere with DNA synthesis and with the ability of PCR polymerases to amplify specific sequences. Thus, UVC-induced damage should correspond to a decrease in PCR signals and an increase in virus inactivation. To test this hypothesis, a PCR assay for PPV was first developed and the sensitivity of the assay was determined. Plasmid pN2Δ contains the complete genome of PPV strain, NADL-2, in a pUC19 background. Aliquots of purified pN2Δ DNA containing 10^5, 10^4, 10^3, 10^2, 10^1 and 10^0 PPV genome copies were amplified, as described in the Materials and methods, using four different sets of primers, to yield four amplicons of different sizes: 0·3 kb, 0·7 kb, 1·2 kb and 2·0 kb. As shown in Fig. 2, all four amplicons shared the same detection limit. The minimum number of genome copies detectable was 10^2 genome copies per PCR reaction, although the intensity of the signal was greater for the 0·3- and 0·7-kb amplicon, than for the 1·2-kb and 2·0-kb amplicons.

The relationship of PPV genome copies to virus infectivity was determined by extracting and amplifying the DNA from a PPV stock. The DNA equivalents from 10^3, 10^2, 10^1, 10^0, 10^{-1} and 10^{-2} $TCID_{50}$ infectious virus particles were amplified using the four different primer pairs, and the results are shown in Fig. 2. As the small 0·3-, 0·7- and 1·2-kb amplicons were detected, even when the DNA equivalent from less than 1 infectious virus particle (10^{-1} $TCID_{50}$) was amplified, the PPV stock must have contained both infectious and non-infectious particles. Detection of the 2·0-kb amplicon required the DNA equivalent from at least one infectious virus particle (10^0 $TCID_{50}$). These results indicate that the PPV stock contained ≈ 100–1000 DNA copies per infectious virus particle and that the detection of the 2·0-kb amplicon corresponded to the presence of infectious virus.

PCR was used to analyse DNA from UVC-treated samples. Alpha$_1$-PI was spiked with the same stock of PPV that was used to develop the PCR assay and the spiked samples were treated by continuous-flow UVC at fluencies ranging from 0

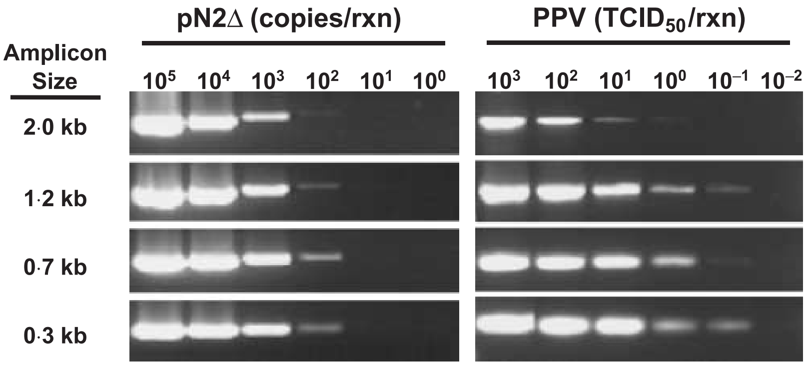

Fig. 2 Detection limit of the porcine parvovirus (PPV) polymerase chain reaction (PCR) assay based on genome copies and virus infectivity. Plasmid pN2Δ, containing the entire PPV genome, was diluted to the specified genome copies and amplified, using four different sets of primers for the 0·3-, 0·7-, 1·2- and 2·0-kb amplicons. Following amplification, the PCR products were separated on agarose gels, which were stained with ethidium bromide (left panel). DNA was extracted from a PPV stock solution and the DNA equivalents from the indicated amount of infectious virus were amplified using the same sets of primers, separated by agarose-gel electrophoresis and stained (right).

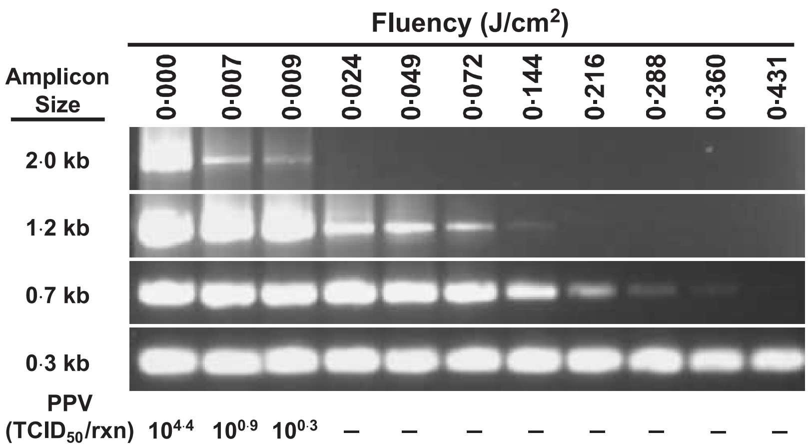

Fig. 3 Effect of ultraviolet C (UVC) irradiation on porcine parvovirus (PPV) DNA and infectivity. PPV-spiked Alpha$_1$-proteinase inhibitor (Alpha$_1$-PI) was irradiated at the specified UVC fluencies and then titrated for residual virus. The DNA, extracted from the treated samples, was amplified using the four sets of primers for the four differently sized amplicons. Following amplification, the polymerase chain reaction (PCR) products were separated and the agarose gels stained with ethidium bromide.

to 0·431 J/cm^2. After treatment, the samples were titrated for residual virus and their DNAs were extracted for PCR amplification. The purified PPV DNA, equivalent to ≈ 100 µl of UVC-treated test sample, was amplified using the four different sets of primers, and the results are shown in Fig. 3. The intensity of the small 0·3-kb amplicon was similar at all fluencies tested, but as the PCR amplicon size increased, the intensity of the signal at higher fluencies decreased. The 0·7- and 1·2-kb amplicons were barely detectable at fluencies of 0·360 J/cm^2 and 0·144 J/cm^2, respectively. Detection of the largest amplicon was, again, associated with the presence of infectious virus, as infectious PPV ($10^{0·3}$ or 2 TCID$_{50}$) and the 2·0-kb amplicon could be detected in Alpha$_1$-PI samples treated with 0·009 J/cm^2 UVC, but were below the detection limit after treatment with 0·024 J/cm^2 UVC. In contrast, UVC treatment did not affect proteins, as recovery of Alpha$_1$-PI activity was 98% after treatment with 0·024 J/cm^2 UVC.

Owing to the difficulty in propagating and assaying B19 *in vitro*, PPV and canine parvovirus (CPV) have been used to model B19 in virus-inactivation studies. Recent experiments, however, have shown that PPV and CPV are more resistant than B19 to thermal inactivation and therefore may not be an appropriate model for B19 in pasteurization studies [15,16]. To test the suitability of PPV as a model for B19 during treatments that affect viral nucleic acids, UVC experiments were performed with PPV and B19. Alpha$_1$-PI spiked with B19-containing human plasma to a final concentration of 10^7 IU/ml, or with PPV to a final concentration of $10^{5·5}$ TCID$_{50}$/ml (≈ $10^{7·5}$ DNA copies/ml), was treated by continuous-flow UVC at various fluencies. Viral DNA was extracted from the UVC-treated samples and amplified using two sets of primers for amplicons that were ≈ 0·3–0·4 kb and 2·0–2·2 kb in length. As shown in Fig. 4, detection of the two B19 amplicons was similar to detection of the PPV amplicons. The 0·3- and 0·4-kb amplicons were present after treatment of B19- or PPV-spiked Alpha$_1$-PI with UVC at fluencies as high as 0·431 J/cm^2, while treatment with fluencies of 0·024 J/cm^2 or higher eliminated detection of the 2·2- and 2·0-kb amplicons. These data suggest that PPV is an appropriate model for B19 in UVC studies.

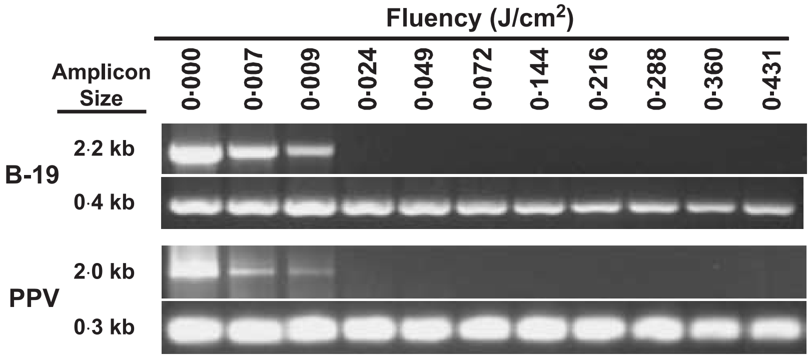

Fig. 4 Comparison of the effect of ultraviolet C (UVC) irradiation on parvovirus B19 (B19) and porcine parvovirus (PPV) DNA. Alpha$_1$-proteinase inhibitor (Alpha$_1$-PI) spiked with B19 (10^7 IU/ml) or with PPV [$10^{5·5}$ tissue culture infective dose 50% ($TCID_{50}$)/ml or ≈ $10^{7·5}$ DNA copies/ml] was irradiated with UVC at the specified fluencies, and the DNA, extracted from the irradiated samples, was amplified to yield the indicated sizes of amplicons. PCR products were separated by agarose-gel electrophoresis and stained with ethidium bromide.

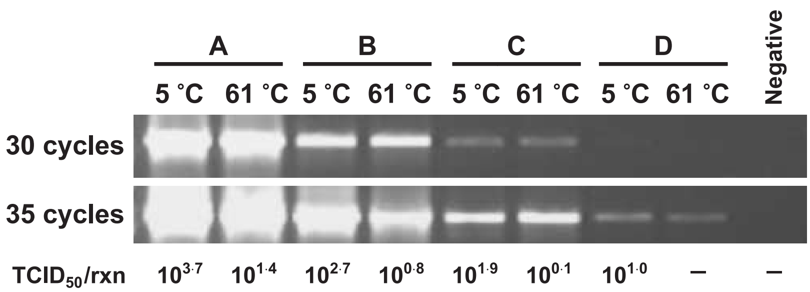

Fig. 5 Effect of wet heat treatment on porcine parvovirus (PPV) DNA and infectivity. Tubes containing 5% albumin, spiked with the indicated concentrations of PPV, were incubated at 61 °C or 5 °C for 20 h. Mock-spiked 5% albumin was incubated in parallel as a negative control. Following incubation, aliquots were removed for virus titration and polymerase chain reaction (PCR) analyses. Primers for the 2·0-kb PPV amplicon were used to amplify the extracted DNA for 30 and 35 cycles. PCR products were separated by agarose-gel electrophoresis and stained with ethidium bromide.

Virus inactivation by heating in solution

Although PCR detection of the 2·2-kb amplicon was a useful indicator for the presence of infectious PPV in UVC-treated Alpha$_1$-PI, PCR assays may not be able to distinguish infectious from non-infectious after treatments that do not affect viral nucleic acids. Pasteurization, or heating in solution at 60 °C for 10 h, inactivates many viruses, presumably by denaturing viral proteins. Experiments, as described in the Materials and methods, were conducted to evaluate the utility of the PCR assay to study virus inactivation during heating in solution. Tubes containing 5% human albumin were spiked with PPV, to final concentrations of 10^5, 10^4, 10^3 and 10^2 $TCID_{50}$/ml, and heated at 61·5 °C for 20 h. As PPV is extremely heat resistant, the excessive temperatures and extended incubation times were necessary to ensure 2·0 $\log_{10}$ PPV inactivation by heating in solution. A duplicate set of PPV-spiked tubes was incubated at 5 °C and served as unheated controls. After treatment, aliquots were removed for virus titration and for PCR analysis using only the primer pairs for the 2·0-kb PPV amplicon. In order to facilitate the comparison between the heated and unheated samples, the extracted DNAs underwent 30 and 35 cycles of PCR amplification. As shown in Fig. 5, ≈ 2 $\log_{10}$ PPV were inactivated. However, the intensity of the PCR signals for the 2·0-kb PPV amplicon were the same for both the heated and unheated samples, at 30 and 35 cycles. These results indicate that heating in solution does not affect viral nucleic acids, so PCR amplification of large amplicons is not an effective method for using to assay virus inactivation during such treatments.

Discussion

A spiral flow UV reactor with novel hydraulic mixing has been designed for uniform and controlled UVC treatment. The current study demonstrates the effectiveness of the Bayer Technology Services UV reactor to inactivate viruses, without the need for quenchers and with minimal damage to the test protein, Alpha$_1$-PI. A diverse panel of viruses (Table 2), including enveloped and non-enveloped viruses with single-stranded or double-stranded, long or short, RNA or DNA genomes, was tested. UVC treatment of Alpha$_1$-PI resulted in over 4 $\log_{10}$ SV40, PPV, HAV, Sindbis, Reo and Adeno inactivation and recovery of greater than 90% Alpha$_1$-PI activity. The kinetics of viral inactivation were relatively linear and no small resistant fraction of virus persisted.

UVC irradiation rarely affects carbohydrates and lipids and usually induces photochemical reactions in heterocyclic bases of nucleic acids and in amino acid residues of proteins and glycoproteins [17]. Under the conditions tested in this study, it is unlikely that viruses were inactivated by UVC-induced changes to proteins, as the highest doses resulted in less than a 10% decrease in $Alpha_1$-PI activity and monomer levels. A PCR assay, developed to assess DNA damage, showed that viral inactivation was caused by UVC-induced damage to viral nucleic acids. The PCR DNA damage detection assay was based on the principle that the DNA damage introduced by UVC inhibited amplification either by blocking primer extension on the DNA template and/or by inhibiting binding of *Taq* polymerase to the template DNA. PPV sequences in a DNA plasmid and in a preparation of infectious virus were amplified using four different sets of primers for amplicons of different sizes. The data showed that all four amplicons shared the same detection limit in the PCR assay, although the intensity of the signal for the smaller amplicons was slightly higher than for the larger amplicons. Detection of the large 2-kb amplicon corresponded to the presence of infectious virus. DNA from PPV- and B19-spiked UVC-treated samples were then amplified using the same primer sets. The signals for the small amplicons were comparable in all UVC-treated samples, regardless of the amount of infectious virus present, and served as internal PCR controls to confirm the presence of the same amount of DNA in each sample. In contrast, detection of the large PPV and B19 amplicons decreased, in a dose-dependent manner, with increasing UVC fluencies, and disappearance of the large PPV and B19 amplicons corresponded to a loss in PPV infectivity.

These results demonstrate that, although PCR-based assays are limited in their ability to discriminate between infectious and non-infectious particles, the PCR assay for large amplicons can be used to estimate the levels of infectious virus present in a sample. In addition, the assay could be adapted to quantify viruses, such as B19, HBV and HCV, which are difficult to propagate *in vitro*. The utility of the large amplicons, as markers for infectious virus, however, would be limited to samples treated with inactivation methodologies that targeted nucleic acids. For example, the assay would not be appropriate to study virus inactivation by pasteurization, which targets viral proteins, so the more labour-intensive and lengthy cell-based assay to measure virus infectivity would be used in these instances.

The results presented here are similar to the reports of studies from other laboratories in showing that doses of UVC irradiation which ensure viral inactivation do not damage proteins [18,19]. It is also generally believed that UV irradiation is especially effective on viruses with large genomes [20] and that viruses with single-stranded genomes are more sensitive to UVC inactivation than viruses with double-stranded genomes [21]. If such assumptions were correct, one of the most UVC-resistant viruses would be the small, double-stranded DNA virus, SV40. However, in the present study, SV40 was the most UVC-sensitive virus. The D_{4L} value for SV40 was lower than that of the single-stranded, RNA-containing Sindbis and the single-stranded, DNA-containing PPV. Adeno and Reo, which were expected to be the easiest to inactivate by UVC treatment owing to their large genomes, were, in fact, the most difficult viruses to inactivate. The unexpected survival curves could not be attributed to differences in the host cell lines used to assay viruses because BSC-1 cells were used to assay SV40, HAV and Reo. Comparison of the results presented here with those of other UVC studies is difficult owing to differences in the shape, size and lamp geometry of UV-irradiation systems and to differences in protein concentrations and compositions of process streams. The discrepancies in results also demonstrate the complexities associated with studies to evaluate UVC inactivation of viruses, as the outcome of such experiments is dependent on the cell systems used to assay UVC-damaged viruses. For example, some viruses may experience UVC-induced DNA damage but appear UVC-resistant because they code for their own repair enzymes [22] or take advantage of host cell repair mechanisms [23]. As PCR is less prone to the variations associated with cell-based assays, and PCR assay conditions are relatively easy to define and control, PCR may be a more reliable method for using to analyse and compare the UVC results from different laboratories. The PCR assay for large amplicons to measure DNA damage may provide a level of biological significance not obtainable with other PCR-based assays. Further studies will be needed to develop the assay as a research tool in quantifying infectious virus.

While B19 infections in healthy individuals may be asymptomatic or benign, they may cause fetal hydrops in women infected during the first half of pregnancy or result in severe chronic anaemia in immunocompromised patients. During epidemic periods, B19 can be detected in the blood of one in 260 donors and B19 viraemia has been induced in recipients of solvent/detergent-treated plasma, intravenous immunoglobulin and factor VIII concentrates [24–26]. Despite careful donor selection, extensive laboratory testing and the introduction of virus inactivation/removal steps in manufacturing processes, a small risk of pathogen transmission by plasma-derived proteins still exists. The present study shows that UVC treatment, in the absence of quenchers, can effectively inactivate viruses, without damaging $Alpha_1$-PI. UVC irradiation inactivated viruses regardless of the type or number of nucleic acid strands, or presence of an envelope. UVC sensitivity was correlated to virus genome size and, based on the correlation, viruses with genomes of < 12 kb in length, such as B19, hepatitis A, B, C, D and G viruses, HIV and human T-cell lymphocytotrophic virus (HTLV), should be inactivated with minimal impact on protein recoveries. In summary, the results are very promising, but additional experiments are necessary to confirm the suitability of UVC-treated $Alpha_1$-PI for use in humans. Moreover, other studies, involving a wide variety of proteins, are needed to assess the general applicability of the new Bayer Technology Services continuous-flow UV reactor.

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